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### GAS INJECTOR ASSEMBLY WITH IMPROVED GAS MIXING

#### Abstract

Gas inserts for semiconductor manufacturing processing chambers with a plurality of injection levels are described. Each of the gas injection levels provides a gas flow to an inner channel within the gas insert. Each of the gas flows are directed in a rotational direction within the inner channel. The gas injection level closest to the outlet end of the gas insert directs a gas flow in the opposite rotational direction to the other gas injection levels. Processing chambers, gas distribution assemblies and methods using the gas inserts are also described.

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#### Background/Summary

##### TECHNICAL FIELD

[0001] Embodiments of the disclosure are directed to gas injectors for semiconductor

manufacturing processing chambers. In particular, embodiments of the disclosure are directed to gas injectors with improved mixing for semiconductor manufacturing processing chambers.

## BACKGROUND

[0002] Reliably producing submicron and smaller features is one of the key requirements of very large scale integration (VLSI) and ultra large scale integration (ULSI) of semiconductor devices. However, with the continued miniaturization of circuit technology, the dimensions of the size and pitch of circuit features, such as interconnects, have placed additional demands on processing capabilities. The various semiconductor components (e.g., interconnects, vias, capacitors, transistors) require precise placement of high aspect ratio features. Reliable formation of these components is critical to further increases in device and density.

[0003] Additionally, the electronic device industry and the semiconductor industry continue to strive for larger production yields while increasing the uniformity of layers deposited on substrates having increasingly larger surface areas. These same factors in combination with new materials also provide higher integration of circuits per unit area on the substrate.

[0004] As the dimensions of devices continue to shrink, so does the gap/space between the devices, increasing the difficulty to physically isolate the devices from one another. Filling in the high aspect ratio trenches/spaces/gaps between devices which are often irregularly shaped with high-quality dielectric materials is becoming an increasing challenge to implementation with existing methods including gap fill, hardmasks and spacer applications.

[0005] Current gas injectors use a cap insert that cannot ensure efficient gas mixing at wide range of process pressure for atomic layer deposition (ALD), or co-flow for chemical vapor deposition (CVD) processes. Multiple tangential and gas entry ports are used to create a vortex inside the cap insert during gas delivery. Current designs result in an entry port signature recognizable in the deposited films due to the gas mixing depending on the types of process being used.

[0006] Accordingly, there is a need in the art for improved gas mixing for gas injectors of processing chambers.

## SUMMARY

[0007] One or more embodiments of the disclosure are directed to gas inserts for semiconductor manufacturing processing chambers. The gas inserts include a plurality of gas injection levels. Each gas injection level provides a gas flow to an inner channel within the gas insert. Each of the gas flows are directed in a rotational direction within the inner channel. The gas injection level closest to an outlet end of the gas insert directs a gas flow in an opposite rotational direction than the other gas injection levels.

[0008] Additional embodiments of the disclosure are directed to gas inserts for semiconductor manufacturing processing chambers. The gas inserts include an inlet end having an inlet end wall with a plurality of openings extending therethrough, and an outlet end having an outlet end face. The inlet end and outlet end define a length of the gas insert. An inner channel extends from the inlet end wall to the outlet end with an opening in the outlet end face. The inner channel is bounded by a channel wall. The gas inserts include a plurality of gas injection levels. Each gas injection level includes a peripheral recess in an outer peripheral wall of the gas insert. The plurality of gas injection levels includes an inlet end gas injection level and an outlet end gas injection level. Each of the peripheral recesses have a plurality of angled apertures extending from a bottom surface of the peripheral recess to the inner channel.

[0009] Further embodiments of the disclosure are directed to gas injector assemblies for semiconductor manufacturing processing chambers. The assemblies include a showerhead, a backing plate, a gas insert and a gas manifold. The showerhead has a front surface and a back surface with a plurality of apertures extending therethrough. The backing plate has a front surface with a concave portion and a back surface with an opening extending to the concave portion of the front surface. The backing plate is positioned adjacent the back surface of the showerhead to form a plenum between the concave portion of the front surface and the back surface of the showerhead.

The gas insert has an outlet end in contact with the back surface of the backing plate. The gas insert includes a plurality of gas injection levels, each gas injection level providing a gas flow to an inner channel within the gas insert. Each of the gas flows are directed in a rotational direction within the inner channel. The gas injection level closest to an outlet end of the gas insert directs a gas flow in an opposite rotational direction than the other gas injection levels. The inner channel is in fluid communication with the plenum through the back surface of the backing plate. The gas manifold is around the gas insert. The gas manifold cooperatively interacts with the gas insert to form peripheral recesses for the gas injection levels.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0010] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

[0011] FIG. 1 shows a cross-sectional schematic view of a processing chamber in accordance with one or more embodiment of the disclosure;

[0012] FIG. 2 shows an expanded view of a gas distribution assembly with a gas insert according to one or more embodiment of the disclosure;

[0013] FIG. 3 shows a side view of a gas insert according to one or more embodiment of the disclosure, the internal structure of which is shown in dotted lines;

[0014] FIG. 4 shows a cross-sectional view of the gas insert of FIG. 3 taken through the center of the inner channel;

[0015] FIG. 5 shows a cross-sectional view of the gas insert of FIG. 3 taken along line 5-5'; and

[0016] FIG. 6 shows a cross-sectional view of the gas insert of FIG. 3 taken along line 6-6'.

### DETAILED DESCRIPTION

[0017] Before describing several exemplary embodiments of the disclosure, it is to be understood that the disclosure is not limited to the details of construction or process steps set forth in the following description. The disclosure is capable of other embodiments and of being practiced or being carried out in various ways.

[0018] As used in this specification and the appended claims, the term "substrate" refers to a surface, or portion of a surface, upon which a process acts. It will also be understood by those skilled in the art that reference to a substrate can also refer to only a portion of the substrate, unless the context clearly indicates otherwise. Additionally, reference to depositing on a substrate can mean both a bare substrate and a substrate with one or more films or features deposited or formed thereon

[0019] A "substrate" as used herein, refers to any substrate or material surface formed on a substrate upon which film processing is performed during a fabrication process. For example, a substrate surface on which processing can be performed include materials such as silicon, silicon oxide, strained silicon, silicon on insulator (SOI), carbon doped silicon oxides, amorphous silicon, doped silicon, germanium, gallium arsenide, glass, sapphire, and any other materials such as metals, metal nitrides, metal alloys, and other conductive materials, depending on the application. Substrates include, without limitation, semiconductor wafers. Substrates may be exposed to a pretreatment process to polish, etch, reduce, oxidize, hydroxylate, anneal, UV cure, e-beam cure and/or bake the substrate surface. In addition to film processing directly on the surface of the substrate itself, in the present disclosure, any of the film processing steps disclosed may also be

performed on an underlayer formed on the substrate as disclosed in more detail below, and the term “substrate surface” is intended to include such underlayer as the context indicates. Thus, for example, where a film/layer or partial film/layer has been deposited onto a substrate surface, the exposed surface of the newly deposited film/layer becomes the substrate surface.

[0020] “Atomic layer deposition” or “cyclical deposition” as used herein refers to a process comprising the sequential exposure of two or more reactive compounds to deposit a layer of material on a substrate surface. “Atomic layer deposition” or “cyclical deposition” as used herein refers to a process comprising the sequential exposure of two or more reactive compounds to deposit a layer of material on a substrate surface. The substrate, or portion of the substrate, is exposed separately to the two or more reactive compounds which are introduced into a reaction zone of a processing chamber. In a time-domain ALD process, exposure to each reactive compound is separated by a time delay to allow each compound to adhere and/or react on the substrate surface and then be purged from the processing chamber. These reactive compounds are said to be exposed to the substrate sequentially. In a spatial ALD process, different portions of the substrate surface, or material on the substrate surface, are exposed simultaneously to the two or more reactive compounds so that any given point on the substrate is substantially not exposed to more than one reactive compound simultaneously. As used in this specification and the appended claims, the term “substantially” used in this respect means, as will be understood by those skilled in the art, that there is the possibility that a small portion of the substrate may be exposed to multiple reactive gases simultaneously due to diffusion, and that the simultaneous exposure is unintended.

[0021] In one aspect of a time-domain ALD process, a first reactive gas (i.e., a first precursor or compound A) is pulsed into the reaction zone followed by a first time delay. Next, a second precursor or compound B is pulsed into the reaction zone followed by a second delay. During each time delay, a purge gas, such as argon, is introduced into the processing chamber to purge the reaction zone or otherwise remove any residual reactive compound or reaction by-products from the reaction zone. Alternatively, the purge gas may flow continuously throughout the deposition process so that only the purge gas flows during the time delay between pulses of reactive compounds. The reactive compounds are alternatively pulsed until a desired film or film thickness is formed on the substrate surface. In either scenario, the ALD process of pulsing compound A, purge gas, compound B and purge gas is a cycle. A cycle can start with either compound A or compound B and continue the respective order of the cycle until achieving a film with the predetermined thickness.

[0022] In an embodiment of a spatial ALD process, a first reactive gas and second reactive gas (e.g., nitrogen gas) are delivered simultaneously to the reaction zone but are separated by an inert gas curtain and/or a vacuum curtain. The substrate is moved relative to the gas delivery apparatus so that any given point on the substrate is exposed to the first reactive gas and the second reactive gas. The gas curtain can be any suitable gas separation arrangement known to the skilled artisan. For example, in some embodiments of a spatial ALD process chamber, a gas curtain is formed by a combination of purge gas ports and vacuum ports to maintain separation between the reactive gases to prevent gas-phase reactions. In some embodiments of a spatial ALD process chamber, separate process stations are configured to form a mini-process environment within each station.

[0023] As used in this specification and the appended claims, the terms “reactive compound”, “reactive gas”, “reactive species”, “precursor”, “process gas” and the like are used interchangeably to mean a substance with a species capable of reacting with the substrate surface or material on the substrate surface in a surface reaction (e.g., chemisorption, oxidation, reduction, cycloaddition). The substrate, or portion of the substrate, is exposed sequentially to the two or more reactive compounds which are introduced into a reaction zone of a processing chamber.

[0024] With reference to FIG. 1, one or more embodiments of the disclosure are directed to a semiconductor manufacturing processing chamber **100**. The semiconductor manufacturing processing chamber **100** comprises a chamber body **101** having sidewalls **102** and a bottom wall

**103** surrounding a chamber interior **105**. The sidewall **102** and bottom wall **103** can be integrally formed or separate component connected together by any suitable connection or fastener known to the skilled artisan.

[0025] The semiconductor manufacturing processing chambers **100** of some embodiments includes a gas distribution assembly **110**. The gas distribution assembly **110** comprises a backing plate **120** and a faceplate **130**. In some embodiments, the processing chamber **100** further comprises a pumping ring **140**. In some embodiments, the pumping ring **140** is considered a separate part from the gas distribution assembly **110**.

[0026] Chamber body **101**, in conjunction with the gas distribution assembly **110** encloses the chamber interior **105** of the semiconductor manufacturing processing chamber **100**. During processing, the chamber interior **105** of the semiconductor manufacturing processing chamber **100** is typically maintained at a controlled pressure (usually a low-pressure environment) using one or more gas inlet (not shown) and one or more exhaust (not shown). The skilled artisan will be familiar with the general construction of the chamber body **101** and the use of gas inlets and exhaust systems.

[0027] The backing plate **120** has a front surface **121** and a back surface **122** that define a thickness of the backing plate **120**. The backing plate **120** has an inner portion **124** and an outer portion **125**. The backing plate **120** contacts the faceplate **130** at the outer portion **125**.

[0028] The backing plate **120** has an inlet opening **123** in a center thereof. The inlet opening **123** extends through the thickness of the backing plate **120** from the back surface **122** to the front surface **121**. The central axis of the backing plate **120** is defined at the center of the inlet opening **123**. The outer peripheral edge of the inner portion **124** of the front surface **121** of some embodiments is concentric with the inlet opening **123**. While the backing plate **120** of some embodiments has an oblong or non-symmetrical shape, the central axis remains at the center of the inlet opening **123** even if that is not the center of mass of the backing plate **120**.

[0029] The front surface **121** of the backing plate **120** at the inner portion **124** has a concave shape. The concave shape of some embodiments has a linear slope from the inlet opening **123** to the outer peripheral edge of the inner portion **124** at the transition to the outer portion **125**, as illustrated in the Figures. In some embodiments, the concave shape has a curved profile from the inlet opening **123** to the outer peripheral edge of the inner portion **124**.

[0030] The gas distribution assembly **110** includes a faceplate **130**, which may also be referred to as a “showerhead”. The faceplate **130** has a front surface **131** and a back surface **132** defining a thickness of the faceplate **130**. The faceplate **130** has an inner portion **133** and an outer portion **134**. The inner portion **133** of the faceplate **130** aligns with the inner portion **124** of the backing plate **120** and the outer portion **134** of the faceplate **130** aligns with the outer portion **125** of the backing plate **120**. The inner portion **133** of the faceplate **130** comprises a plurality of apertures **135** extending through the thickness of the faceplate **130**.

[0031] The backing plate **120** can be connected to the faceplate **130** by any suitable mechanism known to the skilled artisan. For example, the backing plate **120** can be welded to the faceplate **130**. In some embodiments, the backing plate **120** is connected to the faceplate **130** with a plurality of fasteners. Suitable fasteners include, but are not limited to, bolts with or without O-rings.

[0032] When the front surface **121** of the outer portion **125** of the backing plate **120** is in contact with the outer portion **134** of the back surface **132** of the faceplate **130**, a gas box plenum **129** is formed in the space between the front surface **121** of the inner portion **124** of the backing plate **120** and the inner portion **133** of the back surface **132** of the faceplate **130**.

[0033] In some embodiments, the gas box plenum **129** has a coating to improve chemical compatibility. In some embodiments, the coating covers the entire front surface **121** of the backing plate **120** and the entire back surface **132** of the faceplate **130**, including in the inlet opening **123** of the backing plate **120** and the plurality of apertures **135** of the faceplate **130**. In some embodiments, the coating is only on the portions of the backing plate **120** and faceplate **130** that will come into

contact with the process gases.

[0034] In some embodiments, the gas distribution assembly **110** further comprises a gas manifold **195** connected to the back surface **122** of the backing plate **120**. The gas manifold **195** has a gas insert **200** with an inner channel **210** aligned with the opening **123** in the center of the backing plate **120**. The inner channel **210** of some embodiments has an upper portion **202** and a lower portion **203**. The upper portion **202** has a larger inner diameter than the inner diameter of the lower portion **203**.

[0035] Conventional gas distribution assemblies either do not incorporate a gas insert or use a gas insert that suffers from insufficient gas mixing. During deposition, the non-uniform gas mixing is revealed as an entry port signature in the deposited film. The gas insert of the present disclosure advantageously provides efficient gas mixing at a wide range of process pressures for ALD processes, and improved co-flow in CVD processes.

[0036] Some embodiments of the disclosure improve gas mixing with two or more gas injection levels with tangential entry ports with a first rotation direction (e.g., clockwise or anti-clockwise), and one or more bottom gas injection level with tangential entry ports configured to provide a flow of gas in a second direction opposite the first direction. The inventor has found that the addition of the lower injection level generates significant turbulence and acts as a gas curtain to increase the residence time of the gases mixing from the other injection levels before entering the gas box plenum **129**. The bottom injection level can be configured to supply sufficient kinetic energy to break the laminarity of the precursor delivery from the upper injection levels and improve mixing efficiency and uniformity.

[0037] Accordingly, some embodiments are directed to gas inserts for semiconductor manufacturing processing chambers. The gas insert comprises a plurality of gas injection levels, each gas injection level providing a gas flow to an inner channel within the gas insert. Each of the gas flows are directed in a rotational direction within the inner channel. The gas injection level closest to the outlet end of the gas insert directs a gas flow in an opposite rotation direction than the other gas injection levels. The rotational direction of the gas flows within the inner channel causes gases to flow around the longitudinal axis of the gas insert which extends from the inlet end to the outlet end of the gas insert.

[0038] FIG. 2 shows an expanded view of a gas distribution assembly **110** with a gas insert **200** according to one or more embodiment of the disclosure. FIG. 3 shows a side view of a gas insert **200** according to one or more embodiment of the disclosure, the internal structure of which is shown in dotted lines. FIG. 4 shows a cross-sectional view of the gas insert **200** of FIG. 3 taken through the center of the inner channel **210**. FIG. 5 shows a cross-sectional view of the gas insert **200** of FIG. 3 taken along line 5-5'. FIG. 6 shows a cross-sectional view of the gas insert **200** of FIG. 3 taken along line 6-6'.

[0039] With reference to FIGS. 2 through 6, one or more embodiments of the disclosure are directed to gas inserts **200** for a semiconductor manufacturing processing chamber **100**. The gas insert **200** comprises a plurality of gas injection levels **220**, **230**, **240** providing a gas flow to an inner channel **210** within the gas insert **200**. Each of the gas flows are directed in a rotational direction within the inner channel **210**, with the gas level closest to the outlet end **205** of the gas insert **200** directs a gas flow in an opposite rotational direction than the other gas injection levels.

[0040] The embodiments illustrated in the Figures have three gas injection levels. However, the skilled artisan will recognize that there can be more than three gas injection levels with at least the gas injection level closest to the outlet end **205** of the gas insert **200** directing a gas flow in the opposite rotational direction from at least one of the gas injection levels above. In some embodiments, the gas injection level closest to the outlet end **205** of the gas insert **200** directs a gas flow in a direction opposite the other gas injection levels. In some embodiments, at least two gas injection levels direct gas flows opposite at least two other gas injection levels. For example, in an alternating arrangement, or stacked with the at least two opposite flow injection levels closest to the

outlet end **205**.

[0041] In some embodiments, the plurality of gas injection levels comprises an inlet end gas injection level **220** closest to the inlet end **204** of the gas insert **200** and the gas injection level **240** closest to the outlet end **205** of the gas insert **200**. The inlet end **204** and outlet end **205** defining a length of the gas insert **200**.

[0042] In the illustrated embodiment, there are three gas injection levels: an inlet end gas injection level **220** closest to the inlet end **204** of the **200**, an intermediate gas injection level **230**, and an outlet end gas injection level **240**. The outlet end gas injection level **240** is closest to the outlet end **205** of the gas insert **200**. Stated differently, the outlet end gas injection level **240** is furthest from the inlet end **204** of the gas insert **200**.

[0043] The gas insert **200** has an outer peripheral surface **207** that, in combination with the gas manifold **195** creates a plurality of injection level recesses. In the illustrated embodiment, a plurality of peripheral recesses is formed in the outer peripheral surface **207** of the gas insert **200** so that when the gas insert **200** is within the gas manifold **195**, a plurality of injection level recesses is formed. Stated differently, the gas manifold **195** is around the gas insert **200**, and the gas manifold **195** cooperatively interacts with the gas insert **200** to form peripheral recesses of the gas injection levels. The skilled artisan will recognize the complementary nature of the gas insert **200** within the gas manifold **195** and the injection level recesses can be formed on the inner surface of the gas manifold **195** so that the injection levels are formed upon assembly with the gas insert **200**.

[0044] In the illustrated embodiment, each gas injection level **220**, **230**, **240** comprises a peripheral recess **222**, **232**, **242** in the outer peripheral surface **207** of the gas insert **200**. Each peripheral recess **222**, **232**, **242** comprises a plurality of angled apertures **225**, **235**, **245** extending from a bottom surface **223**, **233**, **243** of the respective peripheral recess **222**, **232**, **242** to the inner channel **210**. Each of the angled apertures **225**, **235**, **245** have an outer opening **226**, **236**, **246** as the bottom surface **223**, **233**, **243** of the respective peripheral recess **222**, **232**, **242**, and an inner opening **227**, **237**, **247** at the channel wall **211** of the inner channel **210**.

[0045] The number of angled apertures **225**, **235**, **245** in each of the peripheral recesses **222**, **232**, **242** can affect the gas flow and mixing efficiency within the **210**. In some embodiments, the inlet end gas injection level **220** comprises in the range of 4 to 10 angled apertures **225** spaced around the bottom surface **223** of the peripheral recess **222**. In some embodiments, the intermediate gas injection level **230** comprises in the range of 4 to 10 angled apertures **235** spaced around the bottom surface **233** of the peripheral recess **232**. In some embodiments, there is the same number of angled apertures **225** in the inlet end gas injection level **220** as the number of angled apertures **235** in the intermediate gas injection level **230**. In some embodiments, there are a different number of angled apertures **225** in the inlet end gas injection level **220** than angled apertures **235** in the intermediate gas injection level **230**.

[0046] The angled apertures **225** are illustrated as being evenly spaced around the bottom surface **223** of the inlet end gas injection level **220** and the angled apertures **235** are evenly spaced around the bottom surface **233** of the intermediate gas injection level **230**. However, the skilled artisan will recognize that the distribution of angled apertures is not limited to being evenly spaced. The cross-sectional view of FIG. 5 taken along line 5-5' of FIG. 3 could also be considered the cross-sectional view through the intermediate gas injection level **230** if the number and angles of the angled apertures are the same for the two levels. In some embodiments, the angled apertures **235** of the intermediate gas injection level **230** are vertically aligned with the angled apertures **225** of the inlet end gas injection level **220** so that cross-sectional views through the inlet end gas injection level **220** and intermediate gas injection level **230** would look the same. In some embodiments, the angled apertures of the intermediate gas injection level **230** as rotated around the longitudinal axis of the gas insert **200** (extending through the center of the inner channel **210** from the inlet end **204** to the outlet end **205**). In some embodiments, the angled apertures of the intermediate gas injection level **230** are collectively rotated around the longitudinal axis by an amount about one half the

spacing of the apertures in the inlet end gas injection level **220**. For example, in the illustrated embodiment, there are six angled apertures evenly spaced (every 60°) around the inner channel **210**, and the six angled apertures of the intermediate gas injection level **230** are evenly spaced (every 60°) around the inner channel **210** but offset from the angled apertures of the inlet end gas injection level **220** by 30°. The skilled artisan will recognize that this is merely one possible configuration and that other configurations are within the scope of the disclosure. For example, if there are three gas injection levels above the outlet end gas injection level **240**, and each of the three gas injection levels have six evenly spaced openings, each of the three gas injection levels could be offset by 20° from each other.

[0047] In some embodiments, the outlet end gas injection level **240** comprises in the range of 6 to 24 angled apertures **245** spaced around the bottom surface **243** of the peripheral recess **242**. In some embodiments, there are more angled apertures **245** in the outlet end gas injection level **240** than either of the inlet end gas injection level **220** or intermediate gas injection level **230**. In some embodiments, the outlet end gas injection level **240** has greater than or equal to 1.5×, 1.75×, 2×, or 2.5× of the number of angled apertures in the inlet end gas injection level **220** and/or the intermediate gas injection level **230**. In the embodiment illustrated, the inlet end gas injection level **220** and intermediate gas injection level **230** each have six angled apertures and the outlet end gas injection level **240** has twelve angled apertures, as can be seen from the cross-sectional views of FIGS. 5 and 6.

[0048] The angle that the apertures direct gas flows into the inner channel **210** can impact the mixing and vortex efficiencies. In some embodiments, the angled apertures **225**, **235**, **245** of one or more of the inlet end gas injection level **220**, intermediate gas injection level **230** or outlet end gas injection level **240** are angled tangential to channel wall **211** of the inner channel **210**. The skilled artisan will be aware of the geometric nature of a tangent line to the circular cross-section of the inner channel **210**. As used in this specification and the appended claims, an angled aperture is tangential if a longitudinal axis of the aperture intersects or touches the cross-sectional radius of the inner channel **210** at an angle within  $\pm 2^\circ$ ,  $\pm 1^\circ$  or  $\pm 0.5^\circ$  of perpendicular (i.e., 90°). Stated differently, the longitudinal axis of the angled apertures intersects or touches the cross-sectional radius of the inner channel **210** at an angle in the range of 88-92°, or 89-91°, or 89.5-90.5°.

[0049] As illustrated, the angled apertures of the injection levels are angled inwardly toward the inner channel **210** while remaining substantially perpendicular to the longitudinal axis of the inner channel **210**. In some embodiments, at least one of the angled apertures is further angled toward the outlet end **205** and gas insert **200**. In some embodiments, the plurality of angled apertures **225**, **235**, **245** in one or more of the inlet end gas injection level **220**, intermediate gas injection level **230** or outlet end gas injection level **240** are angled toward the outlet end **205** of the gas insert **200** and connect to the inner channel **210** tangential to the channel wall **211**.

[0050] In some embodiments, the angled apertures of the various injection levels have a diameter in the range of 0.25 mm to 5 mm, or in the range of 0.5 mm to 4.5 mm, or in the range of 0.75 mm to 4 mm, or in the range of 1 mm to 3.5 mm, or in the range of 1.5 mm to 3.25 mm, or in the range of 2 mm to 3 mm. In some embodiments, the angled apertures **245** of the outlet end gas injection level **240** have a smaller diameter than the angled apertures **225** of the inlet end gas injection level **220** or the angled apertures **235** of the intermediate gas injection level **230**.

[0051] The gas insert **200** has an inlet end **204** with an inlet end wall **206** with an inlet end face and an inlet end inner channel face within the inner channel **210**, and an outlet end **205** with an outlet end face **209**. In some embodiments, the inlet end wall **206** comprises a plurality of openings **208** extending through the inlet end wall **206**. Stated differently, the inner channel **210** has an inlet end **204** having an inlet end wall **206** with a plurality of openings **208** extending therethrough.

[0052] The inner channel **210** extends from the inner channel face of the inlet end wall **206** to the outlet end **205** with an opening **212** in the outlet end face **209**. The inner channel **210** is bounded by the channel wall **211**.

[0053] In the illustrated embodiments, the inner channel **210** comprises an upper portion **214** and a lower portion **216**. The upper portion **214** extends an upper portion length from the inlet end wall **206** to the lower portion **216**. The upper portion **214** of some embodiments has a substantially uniform inner diameter along the upper portion **214** length. The lower portion **216** of the inner channel **210** has a flared profile, as shown in the Figures, with an increasing diameter from the upper portion **214** to the outlet end face **209**. The transition from the upper portion **214** to the lower portion **216** occurs at the point where the channel wall **211** of the inner channel **210** changes from a uniform inner diameter to an increasing inner diameter. The upper portion **214** and lower portion **216** align with the upper portion **202** and lower portion **203** of the gas insert **200**.

[0054] Referring to FIGS. 2 and 3, in some embodiments, the outer peripheral surface **207** of the gas insert **200** includes one or more peripheral channel **250** formed therein. The one or more peripheral channel **250** is a recessed portion of the outer peripheral surface **207** configured to hold an O-ring **251**. In the embodiment shown, there are four peripheral channels **250** spaced above and below each of the inlet end gas injection level **220**, the intermediate gas injection level **230** and the outlet end gas injection level **240**. The one or more peripheral channel **250** have O-rings **251** that aid in the formation of a fluid-tight seal between the various gas injection levels to minimize or eliminate leakage.

[0055] In some embodiments, as shown in FIGS. 2-4, the inlet end **204** of the gas insert **200** comprises a flange **218** extending outwardly from the outer peripheral surface **207** of the gas insert **200**. In the embodiment illustrated, the flange **218** is part of the inlet end wall **206**.

[0056] In use, the outlet end face **209** of the outlet end **205** of the gas insert **200** is in contact with the back surface **122** of the backing plate **120**. The gas insert **200** can be connected to the backing plate **120** by any suitable fastener or connection type known to the skilled artisan.

[0057] Operation of the gas insert **200** in use is described with respect to the three gas injection level embodiment shown in the Figures. The skilled artisan will understand the operation of the system using a gas insert **200** with a different number of injection levels. A first gas is flowed through inlet line **229** in fluid communication with the inlet end gas injection level **220** through the gas manifold **195**. A second gas is flowed through inlet line **239** in fluid communication with the intermediate gas injection level **230** through the gas manifold **195**. The second gas and the first gas can be the same or different. A third gas is flowed through the inlet line **249** in fluid communication with the outlet end gas injection level **240** through the gas manifold **195**. The third gas can be the same as one or more of the first gas or second gas, or different from both the first gas and second gas. In some embodiments, the third gas is an inert or diluent gas.

[0058] Some embodiments of the semiconductor manufacturing processing chamber **100** further comprise a remote plasma source (RPS) **185** connected to the gas manifold **195**. In use, a plasma generated in the remote plasma source **185** flows through the plurality of openings **208** in the inlet end wall **206** of the gas insert **200** into the inner channel **210** and the gas box plenum **129**. In some embodiments, an inert gas purge line (not shown) is connected to the inner channel **210** of the gas insert **200** (i.e., between the inlet end wall **206** and the inlet end gas injection level **220**) to provide a continuous inert gas purge to prevent back streaming of gases to the remote plasma source **185**. In some embodiments, inclusion of the inert gas purge eliminates the need for an isolation valve through continuous inert gas purge.

[0059] In some embodiments that use a remote plasma source (RPS) **185**, a gas is flowed from the remote plasma source **185** through the plurality of openings plurality of openings **208** into the inner channel **210**, a first gas is flowed into the inlet end gas injection level **220**, a second gas is flowed into the intermediate gas injection level **230** have the same composition, and a third gas is flowed into the outlet end gas injection level **240**, where the reverse flow from the outlet end gas injection level **240** creates turbulence and mixes the gases together. In this configuration, the first gas and second gas (and possibly third gas) can be the same species or different species which may react with the gas from the remote plasma source **185** (if a reactive gas is flowed). The skilled artisan

will recognize the various reactive and non-reactive gas flow streams possible with the various embodiments.

[0060] Referring again to FIG. 1, the semiconductor manufacturing processing chamber **100** comprises a substrate support **170** within the chamber interior **105**. The substrate support **170** of some embodiments comprises a support body **171** positioned on a support shaft **172**. The support body **171** has a support surface **173** configured to support a semiconductor wafer **108** for processing. The support shaft **172** of some embodiments is configured to move the support body **171** closer to/further from the faceplate **130** and/or around a rotational axis **175** of the support shaft **172**. During processing, the support surface **173** is spaced from the front surface **131** of the faceplate **130** to form a process gap **109**.

[0061] In some embodiments, the support body **171** includes a thermal element **174** configured to heat the semiconductor wafer **108** on the support surface **173**. The thermal element **174** can be any suitable heating mechanism known to the skilled artisan. For example, in some embodiments, the thermal element **174** comprises a resistive heating element that is connected to a power supply (not shown) configured to apply power to the thermal element **174** to heat the support body **171**. In some embodiments, the support body **171** includes an electrostatic chuck (ESC) (not shown). The skilled artisan will be familiar with the construction of the ESC and the manner in which the ESC is powered and employed.

[0062] In some embodiments, as shown in FIG. 1, the semiconductor manufacturing processing chamber **100** includes a radio-frequency (RF) shield **150**. The RF shield **150** is a generally ring-shaped component that is positioned within the interior **105** of the semiconductor manufacturing processing chamber **100** between the substrate support **170** and the sidewall **102**. The RF shield **150** surrounds the support surface **173** of the substrate support **170** and helps to prevent reactive gases from flowing from the process gap **109** to the interior **105** of the chamber body **101**.

[0063] The RF shield **150** has a top end and a bottom end. The top end of some embodiments has a sloped surface configured to direct a gas flow toward the pumping ring **140**. In some embodiments, the top end of the RF shield **150** has a top end surface that is coplanar with the support surface **173** of the substrate support **170**. In some embodiments, where the top end surface of the RF shield **150** is sloped, as shown in the Figures, the highest point of the top end surface is coplanar with the support surface **173** of the substrate support **170**. In some embodiments, the top end of the RF shield **150** has a top end surface that is below the level of the support surface **173**.

[0064] A pumping ring **140** is positioned on a top surface of the choke plate **160**. The pumping ring **140** has a front surface and a back surface defining a thickness of the pumping ring **140**. In use, the back surface of the pumping ring **140** is positioned adjacent to or in contact with the front surface **131** of the faceplate **130**. In some embodiments, in use, the front surface of the pumping ring **140** is positioned in contact with the top surface of the choke plate **160**.

[0065] The pumping ring **140** of some embodiments comprises a vacuum plenum configured to remove process gases from an interior of the processing chamber. The vacuum plenum is formed by the recess in the front surface of the pumping ring **140** when the front surface of the pumping ring **140** is adjacent another surface. For example, as shown in FIG. 1, when the pumping ring **140** is positioned so that the front surface is adjacent to or in contact with the choke plate **160** or chamber sidewall **102**, a pumping volume **145** is formed.

[0066] In some embodiments, the pumping ring **140** is connected to the backing plate **120** with a plurality of fasteners (not shown) that extend through the faceplate **130**. In some embodiments, bolting the backing plate **120** to the pumping ring **140** sandwiches the faceplate **130** between the backing plate **120** and the pumping ring **140**.

[0067] In some embodiments, at least one aperture **146** extends between the recess **143** in the front surface of the pumping ring **140** and the back surface **142** of the pumping ring **140**. In some embodiments, the at least one aperture **146** extends between the recess **143** in the front surface of the pumping ring **140** and an inner face of the pumping ring **140**. The at least one aperture **146** has

a radius equal to a radius of the front surface opening of the angled openings in the faceplate **130**. [0068] During use, the backing plate **120**, faceplate **130** and pumping ring **140**, in addition to other components, may be separated by one or more O-rings to help maintain a fluid-tight seal for the processing chamber. In some embodiments, the gas distribution assembly **110** includes a plurality of O-rings positioned between the backing plate **120** and the faceplate **130** and/or a plurality of O-rings positioned between the faceplate **130** and the pumping ring **140**. In some embodiments, the pumping ring **140** is connected to the choke plate **160** with at least one O-ring positioned between. [0069] Reference throughout this specification to “one embodiment,” “certain embodiments,” “one or more embodiments” or “an embodiment” means that a particular feature, structure, material, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, the appearances of the phrases such as “in one or more embodiments,” “in certain embodiments,” “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily referring to the same embodiment of the disclosure. Furthermore, the particular features, structures, materials, or characteristics may be combined in any suitable manner in one or more embodiments.

[0070] Although the disclosure herein has been described with reference to particular embodiments, those skilled in the art will understand that the embodiments described are merely illustrative of the principles and applications of the present disclosure. It will be apparent to those skilled in the art that various modifications and variations can be made to the method and apparatus of the present disclosure without departing from the spirit and scope of the disclosure. Thus, the present disclosure can include modifications and variations that are within the scope of the appended claims and their equivalents.

## Claims

1. A gas insert for a semiconductor manufacturing processing chamber, the gas insert comprising: a plurality of gas injection levels, each gas injection level providing a gas flow to an inner channel within the gas insert, each of the gas flows directed in a rotational direction within the inner channel, the gas injection level closest to an outlet end of the gas insert directs a gas flow in an opposite rotational direction than the other gas injection levels.
2. The gas insert of claim 1, wherein the plurality of gas injection levels comprising an inlet end gas injection level closest to the inlet end, and the gas injection level closest to the outlet end.
3. The gas insert of claim 1, wherein there are three gas injection levels: an inlet end gas injection level closest to the inlet end of the gas insert, an intermediate gas injection level, and an outlet end gas injection level, the outlet end gas injection level furthest from the inlet end of the gas insert.
4. The gas insert of claim 3, wherein each gas injection level comprises a peripheral recesses in an outer peripheral wall of the gas insert and a plurality of angled apertures extending from a bottom surface of the peripheral recess to the inner channel.
5. The gas insert of claim 4, wherein the inlet end gas injection level comprises in the range of 4 to 10 angled apertures spaced around the bottom surface of the peripheral recess.
6. The gas insert of claim 4, wherein the intermediate gas injection level comprises in the range of 4 to 10 angled apertures spaced around the bottom surface of the peripheral recess.
7. The gas insert of claim 4, wherein the outlet end gas injection level comprises in the range of 6 to 24 angled apertures spaced around the bottom surface of the peripheral recess.
8. The gas insert of claim 4, wherein the plurality of angled apertures in one or more of the inlet end gas injection level, intermediate gas injection level or outlet end gas injection level are angled tangential to the inner channel.
9. A gas insert for a semiconductor manufacturing processing chamber, the gas insert comprising: an inlet end having an inlet end wall with a plurality of openings extending therethrough; an outlet end having an outlet end face, the inlet end and outlet end defining a length of the gas insert; an

inner channel extending from the inlet end wall to the outlet end with an opening in the outlet end face, the inner channel bounded by a channel wall; a plurality of gas injection levels, each gas injection level comprising a peripheral recess in an outer peripheral wall of the gas insert, the plurality of gas injection levels comprising an inlet end gas injection level and an outlet end gas injection level, each of the peripheral recesses having a plurality of angled apertures extending from a bottom surface of the peripheral recess to the inner channel.

**10.** The gas insert of claim 9, wherein the inner channel comprises an upper portion and a lower portion, the upper portion extending an upper portion length from the inlet end wall, the upper portion having a substantially uniform diameter along the upper portion length, the lower portion having a flared profile with an increasing diameter from the upper portion to the outlet end face.

**11.** The gas insert of claim 10, wherein the plurality of angled apertures in each of the gas injection levels connects to the inner channel in the upper portion.

**12.** The gas insert of claim 9, wherein there are three gas injection levels: an inlet end gas injection level closest to the inlet end of the gas insert, an intermediate gas injection level, and an outlet end gas injection level, the outlet end gas injection level furthest from the inlet end of the gas insert.

**13.** The gas insert of claim 12, wherein each of the inlet end gas injection level and intermediate gas injection level independently comprise in the range of 4 to 10 angled apertures spaced around the bottom surface of the peripheral recess.

**14.** The gas insert of claim 12, wherein the outlet end gas injection level comprises in the range of 6 to 24 angled apertures spaced around the bottom surface of the peripheral recess.

**15.** The gas insert of claim 14, wherein the outlet end gas injection level comprises at least  $2\times$  the number of angled apertures than either the inlet end gas injection level or intermediate gas injection level.

**16.** The gas insert of claim 12, wherein the plurality of angled apertures in one or more of the inlet end gas injection level, intermediate gas injection level or outlet end gas injection level are angled tangential to the inner channel.

**17.** The gas insert of claim 12, wherein the plurality of angled apertures in one or more of the inlet end gas injection level, intermediate gas injection level or outlet end gas injection level are angled toward the outlet end of the gas insert and connect to the inner channel tangential to the channel wall.

**18.** The gas insert of claim 9, wherein the inlet end of the gas insert comprises a flange extending outwardly from an outer surface of the gas insert.

**19.** A gas injector assembly for a semiconductor manufacturing processing chamber comprising: a showerhead having a front surface and a back surface with a plurality of apertures extending therethrough; a backing plate having a front surface with a concave portion and a back surface with an opening extending to the concave portion of the front surface, the backing plate positioned adjacent the back surface of the showerhead to form a plenum between the concave portion of the front surface and the back surface of the showerhead; a gas insert having an outlet end in contact with the back surface of the backing plate, the gas insert comprising a plurality of gas injection levels, each gas injection level providing a gas flow to an inner channel within the gas insert, each of the gas flows directed in a rotational direction within the inner channel, the gas injection level closest to an outlet end of the gas insert directs a gas flow in an opposite rotational direction than the other gas injection levels, the inner channel in fluid communication with the plenum through the back surface of the backing plate; and a gas manifold around the gas insert, the gas manifold cooperatively interacting with the gas insert to form peripheral recesses for the gas injection levels.

**20.** The processing chamber of claim 19, wherein each gas injection level of the gas insert comprises a peripheral recesses in an outer peripheral wall of the gas insert and a plurality of angled apertures extending from a bottom surface of the peripheral recess to the inner channel, the plurality of angled apertures in one or more of the inlet end gas injection level, intermediate gas injection level or outlet end gas injection level are angled tangential to the inner channel.

